The Global Climate and Energy Challenge

PUBPOL 290: Special Topics in Public Policy
Spring 2027
Goldman School of Public Policy at the University of California, Berkeley
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Lectures: GTU 102, Thursdays, 9:00 AM – 12:00 AM

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Please add the course number (PUBPOL 290) in the email's object to get a prompt response.

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Course Description

The Global Climate and Energy Challenge (GCEC) is perhaps the greatest problem of the 21st century: ensuring that people have access to inexpensive and reliable energy — critical for human development — without causing disruptive climate change or unduly compromising health and the environment. This course asks how the tools of economics can help us think critically about the GCEC and the trade-offs it poses. Two ideas anchor the course: that market failures are central to why the GCEC is so hard to solve, and that economics and empirical methods give us the quantitative foundation needed to evaluate policy solutions.

The course is organized around two units that mirror these ideas.

The first unit builds the empirical foundations of the Global Climate and Energy Challenge. Understanding the GCEC — and weighing potential responses to it — requires putting numbers on things: the damages from climate change and air pollution, the costs of abatement, the value of reliable energy access, and the trajectories of energy supply and demand. We will pair economic reasoning with empirical analysis to develop these quantitative building blocks, organized around seven key facts that characterize the GCEC. Lecture topics will include the past, present, and future of energy supply and demand, global climate change, air pollution and its health consequences, and case studies of selected energy technologies such as solar photovoltaics, nuclear power, and unconventional oil and gas. Together these foundations supply the empirical inputs needed for any credible cost-benefit analysis of energy and climate policy.

The second unit turns to economics proper. It examines how market failures — externalities, information frictions, public good problems, and incomplete markets — underpin the GCEC, making it such a difficult problem to solve. We will study these failures in detail and examine the theoretical and practical policy solutions — in both developed and emerging economies — that can address them.

A hallmark of the course is the recognition that it is not possible to fully understand the GCEC solely as an academic exercise. The course will therefore include guest lectures from and moderated discussions with industry leaders, elected officials, policymakers and others whose careers are focused on addressing the GCEC. Simulations and policy engagements are equally integral to the course. They give students space to apply lecture concepts through hands-on exercises — such as an emission trading simulation — go deeper into selected topics from lecture, and actively discuss and debate the conclusions raised throughout the course.

Lectures and Schedule (Tentative)

Unit 1: The Facts of the Global Climate and Energy Challenge

Class 1 (January 21st 2027) – Facts 1–3: Energy, living standards, and unequal consumption

Fact 1: Inexpensive and reliable energy is critical for increasing living standards.
Facts 2 & 3: Energy consumption is unequal and is projected to grow rapidly in emerging economies.

Required Readings:

- Greenstone, Michael. "The Economics of the Global Energy Challenge." *AEA Papers and Proceedings*, 114: 1–30. 2024.
- Greenwood, Jeremy. "Evolving Households: The Imprint of Technology on Life." MIT Press (excerpt as in *Penn Today*). 2019.
- Min, Brian, Zachary P. O'Keeffe, Babatunde Abidoye, Kwawu Mensan Gaba, Trevor Monroe, Benjamin P. Stewart, Kimberly Baugh, and Bruno Sánchez-Andrade Nuño. "Lost in the Dark: A Survey of Energy Poverty from Space." *Joule*, 8(7): 1982–1998. 2024.
- Lee, K., E. Miguel, and C. Wolfram. "Appliance Ownership and Aspirations among Electric Grid and Home Solar Households in Rural Kenya." *American Economic Review: Papers & Proceedings*, 106(5): 89–94. 2016.

Optional Readings:

- Breeden, Aurelien and Josh Holder. "As Europe's Heat Waves Intensify, France Bickers About Air-Conditioning." *New York Times*. Aug 12, 2025.
- Kallis, Giorgos, Federico Demaria, and Giacomo D'Alisa. "Degrowth." *International Encyclopedia of the Social & Behavioral Sciences*, 2nd ed., 24–30. 2015.
- Arsenault, Raymond. "The End of the Long Hot Summer: The Air Conditioner and Southern Culture." *The Journal of Southern History*, Vol. L. 1984.
- Arrow, Kenneth, Partha Dasgupta, Lawrence Goulder, Gretchen Daily, Paul Ehrlich, Geoffrey Heal, Simon Levin, Karl-Göran Mäler, Stephen Schneider, David Starrett, and Brian Walker. "Are We Consuming Too Much?" *Journal of Economic Perspectives*, 18(3): 147–172. 2004.
- Piper, Kelsey. "In Defense of the Washing Machine." *Vox*. 2024.
- *Shocked* (Podcast) Episode 1: Bad Energy.
- Greenstone, Michael. "India's Air Conditioning and Climate Change Quandary." *New York Times*. Oct 26, 2016.
- Walsh, Bryan. "The Worst Type of Poverty: Energy Poverty." *Time*. Oct 11, 2011.
- Lomborg, Bjorn. "The Poor Need Cheap Fossil Fuels." *New York Times*. Dec 3, 2013.
- IEA. "World Energy Outlook: Executive Summary." IEA Report. 2016.
- Barry, Ellen, and Coral Davenport. "Emerging Climate Accord Could Push A/C Out of Sweltering India's Reach." *New York Times*. Oct 12, 2016.
- Davis, Lucas, Paul Gertler, Stephen Jarvis, and Catherine Wolfram. "Air Conditioning and Global Inequality." *Global Environmental Change*, Vol. 69. 2021.
- Worland, Justin. "Why Air Conditioning Is a Life-Saver — and a Danger." *Time*. 2018.

Class 2 (January 28th 2027) – Fact 4: Fossil fuels have historically dominated the energy mix and are expected to remain the dominant source of energy

Required Readings:

- Covert, Thomas, Michael Greenstone, and Christopher R. Knittel. "Will We Ever Stop Using Fossil Fuels?" *Journal of Economic Perspectives*, 30(1): 117–38. 2016.
- Fouquet, Roger. "The Slow Search for Solutions: Lessons from Historical Energy Transitions by Sector and Service." *Energy Policy*, 38(11): 6586–6596. 2010.

Optional Readings:

- U.S. Energy Information Administration. "Electricity Explained." Blog post. 2024.
- Bartik, Alexander W., Janet Currie, Michael Greenstone, and Christopher R. Knittel. "The Local Economic and Welfare Consequences of Hydraulic Fracturing." *American Economic Journal: Applied Economics*, 11(4): 105–55. 2019.
- Jacobsson, Staffan, and Volkmar Lauber. "The Politics and Policy of Energy System Transformation — Explaining the German Diffusion of Renewable Energy Technology." *Energy Policy*, 34(3): 256–276. 2006.

Class 3 (February 4th 2027) – Fact 5: Particulate matter air pollution from fossil fuels is the greatest threat to global health, especially in developing countries

Required Readings:

- Dominici, Francesca, Michael Greenstone, and Cass R. Sunstein. "Particulate Matter Matters." *Science*, 344(6181): 257–259. 2014.
- Ebenstein, Avraham, Maoyong Fan, Michael Greenstone, Guojun He, and Maigeng Zhou. "New Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *PNAS*, 114(39): 10384–10389. 2017.

Optional Readings:

- Greenstone, Michael and Ted Gayer. "Quasi-Experimental and Experimental Approaches to Environmental Economics." *Journal of Environmental Economics and Management*, 57: 21–44. 2009.
- Dockery, Douglas W., C. Arden Pope III, Xiping Xu, John D. Spengler, James H. Ware, Martha E. Fay, Benjamin G. Ferris, and Frank E. Speizer. "An Association Between Air Pollution and Mortality in Six U.S. Cities." *The New England Journal of Medicine*, 329(24). 1993.
- Almond, Douglas, Yuyu Chen, Michael Greenstone, and Hongbin Li. "Winter Heating or Clean Air? Unintended Impacts of China's Huai River Policy." *American Economic Review: Papers & Proceedings*, 99(2): 184–90. 2009.
- Cook, R. Dennis. "Smoking and Lung Cancer." In *R.A. Fisher: An Appreciation*, eds. S. Fienberg and D. Hinkley, New York: Springer-Verlag, 182–191. 1980.
- Freedman, David A. "Statistical Models and Shoe Leather." *Sociological Methodology*, 21: 291–313. 1991.

Class 4 (February 11th 2027) – Fact 6: Fossil fuels cause climate change, which has large and unequal impacts

Required Readings:

- Hsiang, Solomon and Robert E. Kopp. "An Economist's Guide to Climate Change Science." *Journal of Economic Perspectives*, 32(4). 2018.
- Carleton, Tamma and Michael Greenstone. "A Guide to Updating the US Government's Social Cost of Carbon." *Review of Environmental Economics and Policy*, 16(2): 196–218. 2022.
- Cook, Thomas D. and Donald T. Campbell. Chapter 1 (pp. 1–36) "Causal Inference and the Language of Experimentation." In *Quasi-Experimentation: Design and Analysis Issues for Field Settings*. Houghton Mifflin: Boston. 1979.

Optional Readings:

- Barrage, Lint and William Nordhaus. "Policies, Projections, and the Social Cost of Carbon: Results from the DICE-2023 Model." *PNAS*, 121(3). 2024.
- Carleton, Tamma and Solomon M. Hsiang. "Social and Economic Impacts of Climate." *Science*, 353(6304). 2016.
- IPCC. "Climate Change 2021: The Physical Science Basis, Summary for Policymakers." 2021.

Class 5 (February 18th 2027) – Fact 7: The climate arithmetic is cruel

Required Readings:

- Pande, Rohini. "The Climate Crisis Is a Crisis of Inequality." *Science*, 381(6661). 2023.
- Carleton, Tamma, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, Robert E. Kopp, Kelly E. McCusker, Ishan Nath, James Rising, Ashwin Rode, Hee Kwon Seo, Arvid Viaene, Jiacan Yuan, and Alice Tianbo Zhang. "Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits." *The Quarterly Journal of Economics*, 137(4). 2022.

Optional Readings:

- Larsen, Kate, Hannah Pitt, Mahmoud Mobir, Shweta Movalia, Alfredo Rivera, Emma Rutkowski, Marie Tamba, Kelly McCusker, and Trevor Houser. "Rhodium Climate Outlook: Probabilistic Projections of Energy, Emissions, and Global Temperature Rise." New York: Rhodium Group. 2023.
- Schöngart, Sarah, Zebedee Nicholls, Roman Hoffmann, Setu Pelz, and Carl-Friedrich Schleussner. "High-Income Groups Disproportionately Contribute to Climate Extremes Worldwide." *Nature Climate Change*, 15. 2025.
- World Bank Group. "Pakistan Country Climate and Development Report." CCDR Series. 2022.

Unit 2: Policy Solutions

Class 6 (February 25th 2027) – Creating policy solutions with greater benefits than costs (CBA)

Required Readings:

- OECD. *Cost-Benefit Analysis and the Environment: Further Developments and Policy Use*. 2018. Chapters 1, 2 (introduction).
- Greenstone, Michael. "Toward a Culture of Persistent Regulatory Experimentation and Evaluation." In *New Perspectives on Regulation*, 113–127. 2009.

Optional Readings:

- OECD. *Cost-Benefit Analysis and the Environment: Further Developments and Policy Use*. Chapter 11 (distribution). 2018.
- Boardman, Anthony E., David H. Greenberg, Aidan R. Vining, and David L. Weimer. *Cost-Benefit Analysis: Concepts and Practice*, 5th ed. Cambridge: Cambridge University Press (Chapters 1–3, 9–10, 19). 2018.
- Sunstein, Cass. *The Cost-Benefit Revolution*. The MIT Press. 2018.
- Sunstein, Cass. "Is Cost-Benefit Analysis for Everyone?" Chicago Unbound, University of Chicago Law School. 2001.

Class 7 (March 4th 2027) – Policy Solution 1: Raising the costs of fossil fuels by pricing externalities or mandates

Required Readings:

First best:

- Mankiw, N. Gregory. "Smart Taxes: An Open Invitation to Join the Pigou Club." *Eastern Economic Journal*, 35. 2009.
- Aldy, J.E. and R.N. Stavins. "The Promise and Problems of Pricing Carbon: Theory and Experience." *The Journal of Environment & Development*, 21(2): 152–180. 2012.

Second best:

- Fowlie, M., M. Greenstone, and C. Wolfram. "Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program: Research Summary." (see Canvas/bCourses)

Optional Readings:

- Pomerleau, S. and E. Dolan. "Carbon Pricing and Regulations Compared: An Economic Explainer." Niskanen Center Working Paper. 2021.
- Metcalf, Gilbert. "Market-Based Policy Options to Control U.S. Greenhouse Gas Emissions." *Journal of Economic Perspectives*, 23(2). 2009.
- Schmalensee, Richard, and Robert N. Stavins. "The SO_2 Allowance Trading System: The Ironic History of a Grand Policy Experiment." *Journal of Economic Perspectives*, 27(1). 2013.

Class 10 (March 11th) – Policy Solution 2: Subsidizing innovation in policy

ETS Simulation

Required Readings:

- Greenstone, Michael, Rohini Pande, Nicholas Ryan, and Anant Sudarshan. "Can Pollution Markets Work in Developing Countries? Experimental Evidence from India." *The Quarterly Journal of Economics*, 140(2). 2025.
- Duflo, Esther. "The Economist as Plumber." *American Economic Review: Papers & Proceedings*, 107(5). 2017.

Optional Readings:

- Greenstone, Michael, Jesse F. Buchsbaum, and Katherine Meckel. "Can Machine Learning Improve the Environment? Evidence from a National Field Test Targeting Hazardous Waste Inspections." Working paper. 2024.
- Fowlie, Meredith, Michael Greenstone, and Catherine Wolfram. "Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program." *The Quarterly Journal of Economics*, 133(3). 2018.

Class 8 (March 18th 2027) – Midterm

Class 9 (April 1st 2027) – Policy Solution 3: Subsidizing innovation in R&D

Required Readings:

- Nelson, R. "The Simple Economics of Basic Scientific Research." RAND Corporation. 1959.
- Jaffe, A., R. Newell, and R. Stavins. "A Tale of Two Market Failures: Technology and Environmental Policy." *Ecological Economics*, 54(2–3). 2005.
- Popp, D. "R&D Subsidies and Climate Policy: Is There a 'Free Lunch'?" *Climatic Change*, 77: 311–341. 2006. (You may skip Section 2.)

Optional Readings:

- Wolak, Frank. "The End of Expensive Oil?" SIEPR Policy Brief. 2015.
- International Energy Agency (IEA). "R&D and Technology Innovation." *World Energy Investment Report*. 2020. `https://www.iea.org/reports/world-energy-investment-2020/rd-and-technology-innovation`
- Bloom, N., M. Schankerman, and J. Van Reenen. "Identifying Technology Spillovers and Product Market Rivalry." *Econometrica*, 81: 1347–1393. 2013.
- Deutch, John, and Edward Steinfeld. "A Duel in the Sun: The Solar Photovoltaics Technology Conflict between China and the United States." MIT Future of Solar Energy Study Working Paper. 2013.
- Condliffe, James. "Clean Coal's Flagship Project Has Failed." *MIT Technology Review*. June 29, 2017.
- Myers, Kyle R., and Lauren Lanahan. "Estimating Spillovers from Publicly Funded R&D: Evidence from the US Department of Energy." *American Economic Review*, 112(7). 2022.
- Popp, David. "Promoting Innovation for Low-Carbon Technologies." The Hamilton Project. 2019.

- Keith, David. *A Case for Climate Engineering.* The MIT Press. 2013.

Class 11 (April 8th 2027) – Policy Solution 4: Improving access to accurate information

Required Readings:

- Sorin, Jeanne, Sun, Yixin, Resosudarmo, Budy, Suryadarma, Daniel, and Dil Rahut, "Clean Air as a Luxury Experience Good."Working Paper, 2026.
- Shrader, J.G., L. Bakkensen, and D. Lemoine. "Fatal Errors: The Mortality Value of Accurate Weather Forecasts." NBER Working Paper No. 31361. 2023. `https://doi.org/10.3386/w31361`

Optional Readings:

- Greenstone, Michael, and B. Kelsey Jack. "Envirodevonomics: A Research Agenda for an Emerging Field." *Journal of Economic Literature*, 53(1). 2015.
- Resosudarmo, Budy, Jeanne Sorin, and Yixin Sun. "Clean Indoor Air in Jakarta: An Unaffordable Experience." Working Paper. 2025.
- Greenstone, Michael, Guojun He, Ruixue Jia, and Tong Liu. "Can Technology Solve the Principal-Agent Problem? Evidence from China's War on Air Pollution." *American Economic Review: Insights*, 4(1). 2022.
- Greenstone, Michael, Guojun He, Shanjun Li, and Eric Yongchen Zou. "China's War on Pollution: Evidence from the First 5 Years." *Review of Environmental Economics and Policy*, 15(2). 2021.

Class 12 (April 15th 2027) – Policy Solution 5: Moving away from electricity as a right

Required Readings:

- Burgess, Robin, Michael Greenstone, Nicholas Ryan, and Anant Sudarshan. "The Consequences of Treating Electricity as a Right." *Journal of Economic Perspectives*, 34(1). 2020.

Film (∼60 minutes):

- *Aayi Gayi – When Will Electricity Come?* (Anandana Kapur, Documentary, 2019, 61 min.) [link]

Optional Readings:

- "Power to the Powerless." *The Economist.* Feb 17, 2016.
- Josey, Ann, Manabika Mandal, and Shantanu Dixit. "India's Electricity Companies Have Surplus Power — and That's a Big Problem." *Quartz.* Jun 2, 2017.

Class 13 (April 22nd 2027) – Policy Solution 6a: International coordination for climate through negotiations and a system of compensations

Class 14 (April 29nd 2027) – Policy Solution 6b: International coordination for climate through negotiations and a system of compensations

Required Readings:

- Houser, Trevor, Kate Larsen, and Michael Greenstone. "Does the World Free Ride on US Pledges to Reduce Greenhouse Gas Emissions? Evidence from the Paris Climate Agreement." University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2023-146. 2023.

Optional Readings:

- Nordhaus, William. "Climate Clubs: Overcoming Free-Riding in International Climate Policy." *American Economic Review*, 105(4). 2015.
- Lin, Boqiang and Hengsong Zhao. "Threatening the Poor? The Economic Impacts of the Carbon Border Adjustment Mechanism on Developing Countries." *Structural Change and Economic Dynamics*, 71: 582–593. 2024.

Guest Lectures and Discussions

To be updated:

- XXX
- XXX
- XXX
- XXX

Grading Scheme

To be updated

- In class quizzes & attendance
  - We will have a short quiz in class each week, on paper, closed book. The quizzes will cover materials and discussions from the previous lectures, as well as the readings due for that week. The quizzes will be graded very generously. The goal is for you to engage with the material on a regular basis, rather than asking you about details of all the papers we'll have covered.
  - I will take an average of the 10 best quizzes (out of 12).
  - Class attendance is mandatory: if you miss class that day, you'll receive 0 on the quizz.
  - Attendance to the ETS and climate negotiation simulations is mandatory.
- Participation
  - This course works best if you participate. Participation can take many forms, including asking clarifying questions, volunteering to render group discussions, being active in the simulations, sharing relevant articles on the course's *bcourse* etc and engaging with your fellow students in the commentaries.
- Assignments and problem sets
- Exams/final project
  I will distribute practice exam questions beforehand.

Simulations

Emission Trading Simulation

This simulation will take place during class on Date TBD

- Participation is mandatory.
- You will participate in groups.
- You only need to submit one pre-simulation assignment and one post-simulation reflection per group.

More details will be provided closer to the date.

Climate Negotiations Simulation

This simulation will take place during class on Date TBD.

- Participation is mandatory.

- You will participate in groups.
- You only need to submit one pre-simulation assignment and one post-simulation reflection per group.

More details will be provided closer to the date.

Class Policies

1. Late submissions will not be accepted and will result in a score of 0.
2. Re-grade requests: If you request that your assignment (including exams) be regraded, all questions will be reviewed and strictly regraded.
3. Absences: class attendance is mandatory. You are allowed two free pass per semester, no question asked, **except for the ETS and climate negotiation simulations**. Any other absence would have to be justified beforehand.

In-Class Use of Technology

I strongly discourage the use of computers during class. Engaging with the material through active participation and discussion is essential to your learning, and computer use can be a distraction both for you and your peers. Please refrain from using laptops unless absolutely necessary for specific class activities.

The use of cell phones during class is not tolerated under any circumstances. Phones should be silenced and put away before the start of class. If I catch you using a cell phone during class, I will ask you to leave and you will not receive credit for class attendance and participation for the session.

Use of Generative Artificial Intelligence

Please cite any AI tools when/if you use them to complete assignments. If you do not cite, you will receive a 0 for that assignment. I know that AI can do most of your assignments in this class but I am going to tell you where I think you should use AI and what you should not!

Intellectual Property

Course materials (e.g., handouts, exercises, slides) are provided for your personal use. They may not be distributed to others, including by uploading them to external sites (such as CourseHero) or sharing them with individuals who are not GSPP students.

Accommodations

GSPP is committed to fostering an inclusive learning environment that supports the needs of all students, including those with disabilities. If you require an accommodation, please notify me during the first week of class or as soon as your accommodation is finalized with the Disabled Students' Program (DSP).

The DSP Office will determine eligibility and coordinate accommodations. Students are expected to

communicate proactively with the instructor about the need for accommodation in specific situations (in advance of the need, or as soon as practical thereafter). Failure to communicate in a timely way may limit my ability to implement the accommodation.

Compliance with AB 789 (Financial Aid Satisfactory Academic Progress)

To maintain eligibility for financial aid, students must meet Satisfactory Academic Progress (SAP) standards. Satisfactory academic progress requires maintaining progress toward degree completion by completing at least 67% of attempted units, maintaining a minimum cumulative grade point average (2.0 for undergraduate students; 3.0 for graduate students), and completing degree requirements within the maximum allowable timeframe (measured in units for undergraduates and academic terms for graduate students). Failure to meet financial aid SAP standards may result in the suspension of financial aid eligibility unless a SAP appeal is granted. Students with questions regarding their SAP status or the SAP appeals process are encouraged to contact Cal Student Central for assistance.